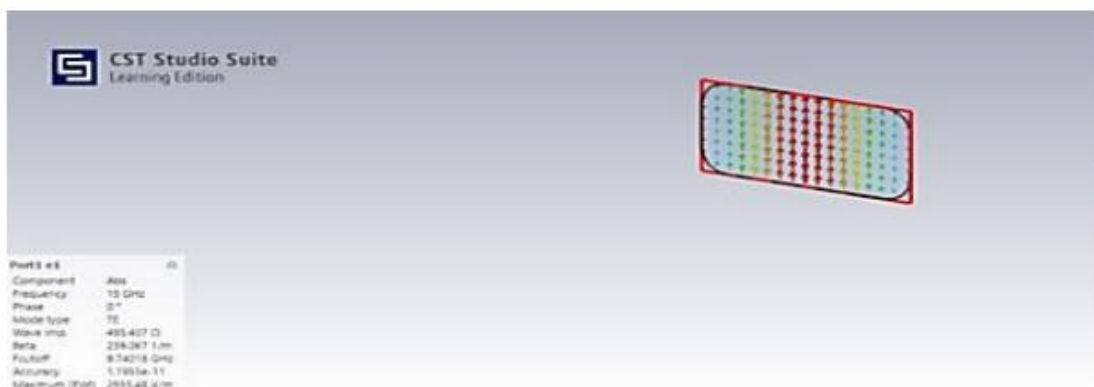
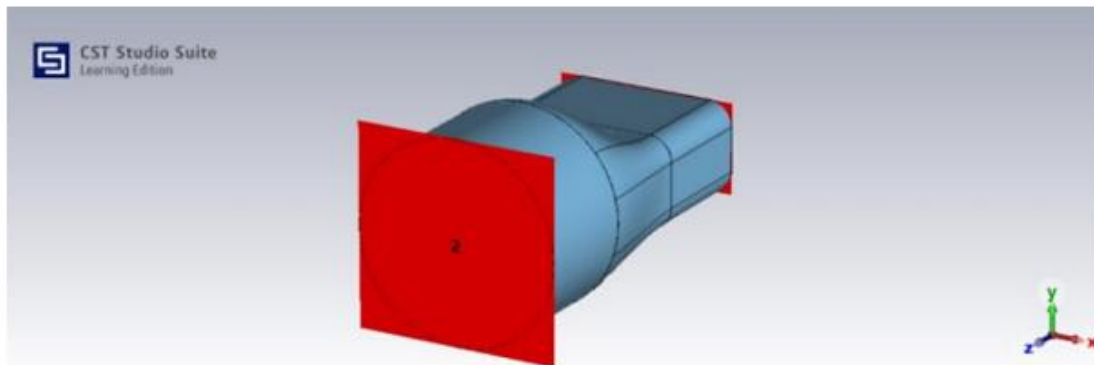


EXNO:13 Investigation of electromagnetic field distribution in rectangular to circular waveguide transition.

Test Case 1: Analyze the electric field and magnetic field distribution at Port 1 of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

Simulation Results:

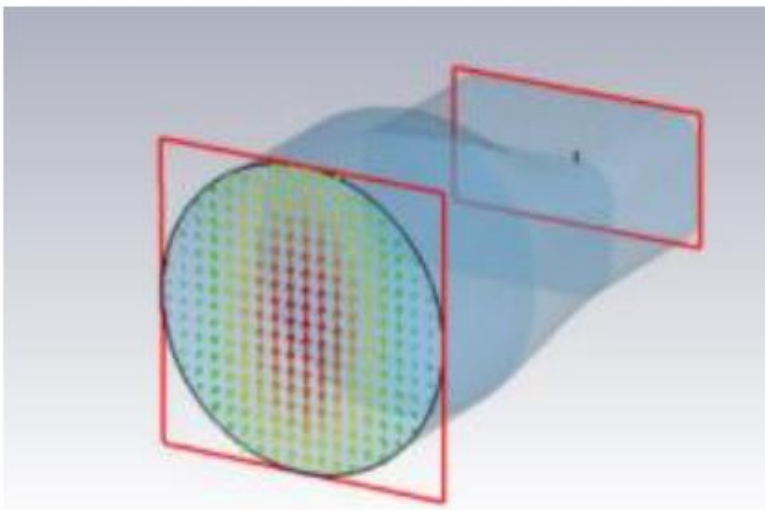


Output:

Parameter	Port 1
e1 (Electric Field)	TE mode
h1 (Magnetic Field)	TE mode
Frequency	15 GHz
Mode type	TE
Wave impedance	$\approx 677 \Omega$
Phase constant β	$\approx 213.5 \text{ 1/m}$
Cutoff frequency	$\approx 8.5 \text{ GHz}$
Peak Value	$\approx 255.4 \text{ V/m}$

Test Case 2: Analyze the electric field and magnetic field distribution at Port 2 of the rectangular waveguide to circular waveguide transition and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

Simulation Result:



Output:

Parameter	Port 2
e2 (Electric Field)	TE mode
h2 (Magnetic Field)	TE mode
Frequency	15 GHz
Mode type	TE
Wave impedance	$\approx 537.3 \, \Omega$
Phase constant β	$\approx 211.6 \, \text{rad/m}$
Cutoff frequency	$\approx 13.8 \, \text{GHz}$
Peak Value	$\approx 2.47 \, \text{kV/m}$